

5 The random variable X has the geometric distribution $\text{Geo}(p)$.

(a) Show that the probability generating function of X is $\frac{pt}{1-qt}$, where $q = 1-p$. [3]

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(b) Use the probability generating function of X to show that $\text{Var}(X) = \frac{q}{p^2}$. [5]

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Kenny throws an ordinary fair 6-sided dice repeatedly. The random variable X is the number of throws that Kenny takes in order to obtain a 6. The random variable Z denotes the sum of two independent values of X .

(c) Find the probability generating function of Z . [2]

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